Fe

Ridge lease R-07: what the block model contains

Cut-off 45% Fe · existing drilling (14 holes) · model seed 2655

All data in this report is synthetic.
The geology, assays, drill holes, prices and costs were generated by a computer model from a fixed random seed to illustrate how a block model works. They do not describe any real place or deposit. This is not a mineral resource or reserve estimate under UNFC, JORC, NI 43-101 or any other reporting code, and must not be presented as the result of field work, sampling or laboratory assays.

Summary

Above the 45% Fe cut-off the model holds 55.3 million tonnes at an average estimated grade of 59.7% Fe, with a tonnage-weighted mean confidence of 0.54. 60.9% of that tonnage is at G2 or better (confidence of 0.5 or more), and 23.1% is G4, ground with almost no drilling nearby.

Resource above cut-off	G1	G2	G3	G4
Million tonnes	19.0	14.6	8.9	12.8

1. Rock types: how much, how rich, how sure

Share is the percentage of all 408.4 Mt of rock in the model. Grade is the tonnage-weighted estimated iron content. Confidence (0 to 0.97) reflects distance to the nearest drill hole. Chance is the average probability that a cell of that rock beats 45% Fe, given its grade uncertainty.

Rock type	Mt	Share	Fe grade	Confidence	Chance
Hematite	34.5	8.4%	62.9%	0.52	97.6%
Blue dust	6.1	1.5%	64.8%	0.53	98.7%
Lateritic ore	11.7	2.9%	50.9%	0.65	82.5%
BHQ (banded hematite quartzite)	147.8	36.2%	36.8%	0.50	15.5%
Waste (phyllite, shale)	208.4	51.0%	14.0%	0.19	–

Share of each rock type's tonnes by confidence stage:

Rock type	G1	G2	G3	G4
Hematite	32.5%	26.4%	13.2%	28.0%
Blue dust	33.6%	29.6%	11.7%	25.1%
Lateritic ore	44.3%	28.9%	26.8%	0.0%
BHQ (banded hematite quartzite)	31.7%	22.1%	16.2%	30.0%
Waste (phyllite, shale)	5.3%	7.9%	23.2%	63.6%

2. Ore above 45% Fe by type of occurrence

Occurrence	Mt	Of resource	Fe grade	Confidence	G2+
Reef ore	30.9	56.0%	63.0%	0.52	59.6%
Pocket deposit	3.5	6.4%	61.9%	0.46	52.2%
Blue dust	6.1	11.1%	64.8%	0.53	63.2%
Lateritic ore	11.1	20.1%	51.3%	0.66	74.2%
Low-grade BHQ	3.5	6.4%	46.1%	0.38	34.5%

Pocket deposits are hematite bodies of fewer than 70 connected cells; larger bodies are reef ore. 143 float-ore fragments are mapped on the slopes below ore outcrops.

3. Grade and tonnage at other cut-offs

Cut-off	Mt	Fe grade	Mt at G2 or better
40% Fe	90.5	52.8%	51.8
45% Fe	55.3	59.7%	33.6
50% Fe	47.1	61.9%	29.2
55% Fe	42.5	62.9%	25.9
58% Fe	39.7	63.3%	24.1
60% Fe	34.3	64.0%	20.7
62% Fe	25.5	65.0%	15.0
64% Fe	16.8	66.0%	9.9

4. Drill holes

Hole	Type	East	North	Depth	Best intercept
DDH-01	Core	125 m	425 m	170 m	40 m at 63.4% Fe from 110 m
DDH-02	Core	250 m	350 m	190 m	40 m at 65.4% Fe from 110 m
RC-03	Reverse circulation	350 m	375 m	160 m	60 m at 60.7% Fe from 0 m
DDH-04	Core	475 m	325 m	200 m	30 m at 56.1% Fe from 0 m
DDH-05	Core	575 m	375 m	200 m	None above 45% Fe
RC-06	Reverse circulation	675 m	275 m	200 m	20 m at 48.6% Fe from 0 m
DDH-07	Core	800 m	275 m	170 m	20 m at 66.7% Fe from 80 m
DDH-08	Core	925 m	250 m	190 m	30 m at 62.1% Fe from 70 m
RC-09	Reverse circulation	200 m	525 m	160 m	30 m at 58.6% Fe from 130 m
DDH-10	Core	425 m	450 m	180 m	10 m at 53.3% Fe from 10 m
DDH-11	Core	725 m	400 m	160 m	70 m at 63.9% Fe from 0 m
RC-12	Reverse circulation	325 m	200 m	160 m	None above 45% Fe
DDH-13	Core	625 m	175 m	170 m	None above 45% Fe
DDH-14	Core	875 m	125 m	160 m	10 m at 45.5% Fe from 30 m

How the numbers are made

- Grid of 44 × 32 × 20 cells, each 25 m × 25 m × 10 m. Tonnes = volume × density (waste 2.6, BHQ 3.3, hematite 4.2, blue dust 3.2, laterite 3.0 t/m³).
- Hematite and blue dust are iron oxide (Fe2O3, about 70% Fe when pure). BHQ is hematite banded with quartz, so its iron content is lower. Laterite is a weathered, iron-rich surface cap.
- Confidence decays with horizontal distance to each hole (exp(-d²/3.3²), zero beyond 7 cells) and below the hole's bottom; the top two cells get at least 0.3 from surface mapping.
- Estimated grade departs further from the model's true grade where confidence is low. Uncertainty sigma = 1.5 + 11 × (1 - confidence) % Fe; chance above cut-off = normal CDF of (grade - cut-off) / sigma.
- Stages follow the spirit of the UNFC geological axis: G1 at confidence 0.75 or more, G2 at 0.5 or more, G3 at 0.25 or more, otherwise G4. Figures use existing drilling only.